

Preparing to Teaching Workshop
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Unpacking the Knapsack

Designing Classrooms and Teaching Practices for Equity

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| Access to these Slides & Internal Links

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| Hopes and Goals

Promote an Equity Mindset

- What are students **bringing** with them to the classroom? *How do I know what they are bringing?*
- What do I want all students to **leave the class** with? *What would success look like?*
- What do students **need to succeed**?
- What **intentional structures** can I use to ensure all students have what they need to succeed?
- What is **success**?!

Introduce Concepts, Examples, & Resources

- Invisible Knapsack
- Backwards & Universal Design
 - Pedagogical choices (e.g. active learning)
 - Course policies (e.g. late policies, test revisions/corrections)
 - Accessible course materials (e.g. price, printed vs. html vs. pdf)
- Alternative grading practices (align with values & pedagogical goals)

| What I bring...

Context before Content

Identities

- White Midwest middle-age woman
- Upper middle class, 3rd generation college educated

Experiences

- Childhood: 4th largest "city" in North Dakota
- Undergraduate: Small Liberal Arts College (SLAC) in Minnesota
- Graduate: Research University in California
- Professional:
 - Assistant Prof. at SLAC in Massachusetts →
 - Asst., Assoc., Full Professor at Macalester College (SLAC) in Minnesota



Me with my three girls! 6 months, 6 years, 10 years

| What I bring...

Context before Content

Statistics/Data Science Teaching

- Over 13 years of post-PhD teaching
- Taught & Developed* Variety of Courses
 - 100-level: **Intro Statistics**, Intro Data Science
 - 200-level: Statistical Machine Learning, Intermediate Data Science*, **Data & Society***
 - 300-level: Multivariate Statistics*, Linear Models, Longitudinal Data Analysis*
 - 400-level: Data Mining, **Correlated Data***, Mathematical Statistics, Projects in Data Science*

But most of what I'm talking about isn't about Statistics...

| The Invisible Knapsack

What are students bringing with them to the classroom?

Unearned Assets

The "**Invisible Knapsack**" represents an unearned package of assets, resources, codes, and maps that privileged students carry automatically (McIntosh 1989).

Hidden Burdens

Students also carry social hurdles that prevent them from engaging as they might desire.

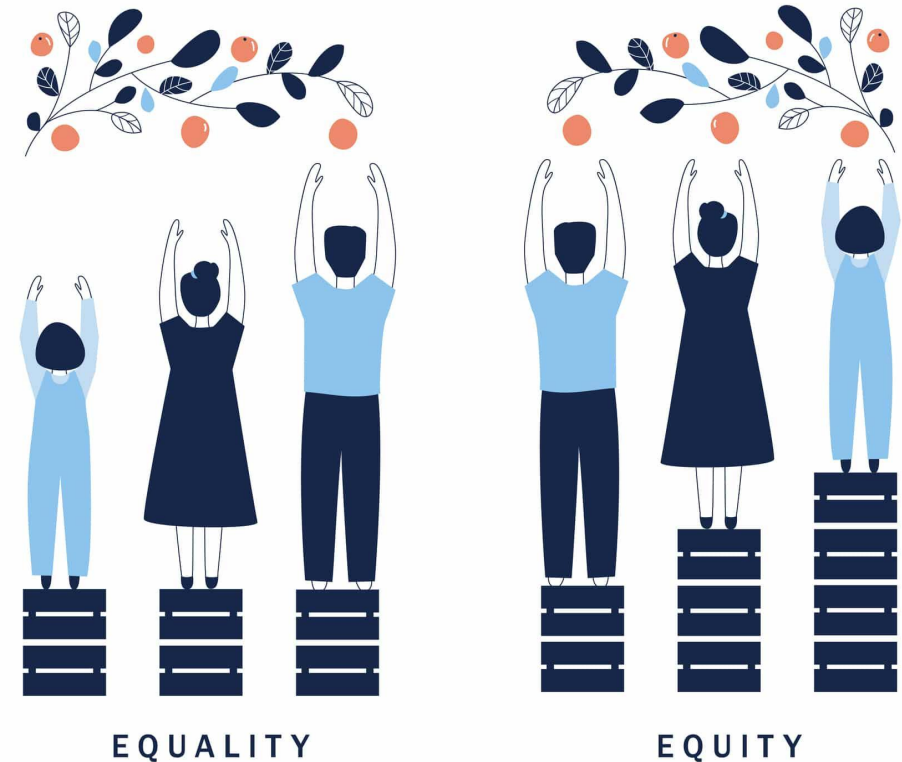


Image Source: Virginia Kindergarten Readiness Program.
<https://vkrponline.org/addressing-equity-in-early-learning/>

| The Knapsack Inventory

Standard Comforts

We may implicitly assume all students walk into our classrooms with similar resources and tools.

- Guaranteed quiet, private space to study and work.
- Constant, high-speed, reliable internet and technology needed to complete the work.
- Representations of their identity within the curriculum.
- Presentation in a language, culture, format they can fully understand.

Unintentional Barriers

Without intentional design, standard policies and pedagogies can serve as traps or barriers.

- Rigid penalties on late work, harsh consequences of unexpected shifts in hourly work.
- Penalizing non-standard dialects as "unprofessional".
- Large group discussions lift up extroverts with social power and confidence.

| The Knapsack Inventory

How do we know what they are bringing?

Self Education

- Educate yourself about experiences different from your own ([books, podcasts, etc.](#))
- Never force students to self-disclose

Cultivate a Culture Safe for Disclosure

- Start of Semester: Information Gathering Survey (opportunities to share concerns about future learning)
- [Mid-course interviews](#) (facilitated by another faculty member)
- [Student Reflections](#) (What is working for their learning? What can they do? What can the instructor do?)
- Anonymous Forms (to report issues or concerns throughout the semester)

| Backwards Design

What do you want students to leave the class with?

Start with the Learning Goals

Step 1: Identify what students should know and be able to do by the end of the unit/course.

Step 2: Design assessments to measure that learning. *Now in an age of AI...*

Step 3: Plan a sequence of lessons & activities that will prepare students to successfully complete the assessments. *Ideas to come...*

| The Hidden Curriculum

What do students need to succeed?

- 🔍 **Making the Implicit Explicit:** Do not assume students "know" how to study, note-take, or read a syllabus. Directly teach critical academic navigation tools.
- 📌 **Demystifying Faculty Expectations:** Explicitly review and explain standard concepts. For example, clarify that "Office Hours" are for drop-in student assistance, not punitive sessions.
- ✅ **The TILT Framework:** Format assignment structures transparently using three precise sections: (1) Purpose, (2) Step-by-Step Task, and (3) Explicit Success Criteria.

Examples

"How To Study" Guide (Intro Stats)

Syllabus (Intro Stats)

Project Guidelines (Intro Stats)

Schedule (Correlated Data)

| Your Turn: Student Guidebook



Basics

Consider a resource, policy, process that is common in higher education but often misunderstood.



Audience

Think about a first generation college student and their familiarity with that "basic" tool.



Guidebook

How would you describe how to navigate that "basic" tool to this audience in your syllabus or course?

Universal Design for Learning

What **structures** ensure all students have what **they need to succeed**?

Designing for the "Exceptions"

Rather than designing for the "mean", design structures that encompass the margins.

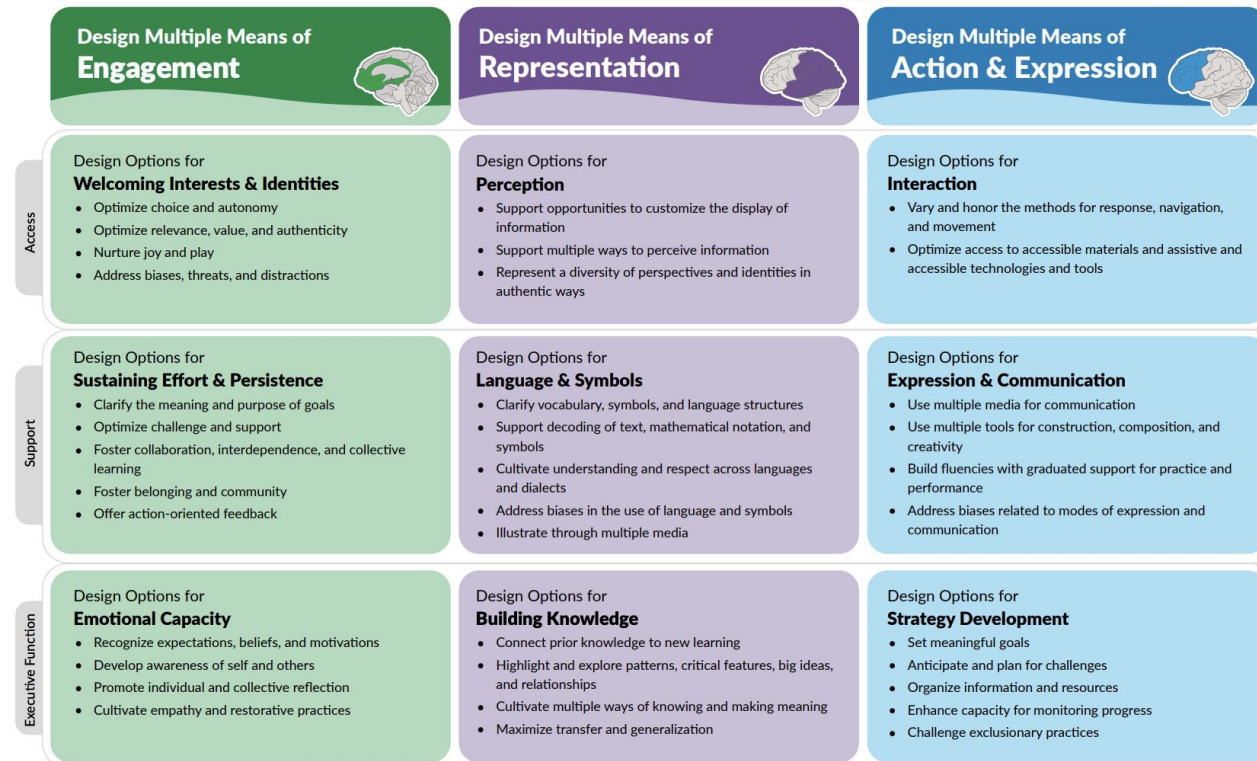
By providing multiple modes of engagement, representation, and expression, we ensure the entire classroom succeeds.

Examples

Accessible Datasets: Avoid data contexts are overly technical or require "hidden" cultural knowledge to interpret

Accessible "Materials": Sharing audio recording of class in private online drive

Supportive Deadlines: Every student gets 3 three-day extensions, no questions asked



| Your Turn: Revision Challenge



The Baseline

Consider a "traditional" class policy that you've experienced as a student or teacher.



Objective

What is the goal or purpose of the policy?



Alternatives

Consider alternatives that still meet that goal but acknowledge different backgrounds and within your professional boundaries.

| Equitable Grading Practices

What is "success"?!

Behavior v. Understanding

Ensure your grading structures calculate actual, verified understanding of the subject matter, rather than assessing a student's environment, compliance, or timing.

Examples

- Eliminate **devastating zeros and extreme penalties** for late submissions that block growth.
- Many **low stakes** assignments with **usable feedback**
- Test corrections/revisions
 - Individual effect
 - Group or collaborative effort (in class or outside of class) -> leads to learning
- Learning reflections (metacognition)
 - Provide additional information about students' perception of their learning

| A Question of Purpose

“When you look at how you grade, what does it reflect about who you are and what you believe? And what do you have to do differently?”

— Grading for Equity

| In the age of AI

Equity is even more important

As you consider your AI policies, consider:

- Non-Native English students
- Past over reliance on AI among students
- Different access and experience with Gen AI products & models
- Recent literature about equity issues with AI
 - Liu, G., Christian, B., Dumbalska, T., Bakker, M. A., & Dubey, R. (2026). AI assistance reduces persistence and hurts independent performance. arXiv. <https://arxiv.org/abs/2604.04721>
 - Prather, J. et al. (2024). The Widening Gap: The Benefits and Harms of Generative AI for Novice Programmers. arXiv. <https://arxiv.org/abs/2405.17739>

Let's Connect

What is one thing you want to try this coming academic year?

Remember: Equity is a mindset.

You'll iterate and adapt to your students over time.

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Resources

Equity & Social Justice

Inclusive Teaching: Strategies for Promoting Equity in the College Classroom by Hogan and Sathy:

<https://wvupressonline.com/inclusive-teaching>

Unpacking the Invisible Knapsack by McIntosh:

<https://www.nationalseedproject.org/key-seed-texts/white-privilege-unpacking-the-invisible-knapsack>

Is Everyone Really Equal? by DiAngelo and Sensoy: https://www.tcpress.com/products/is-everyone-really-equal_9780807787786

Design

Understanding by Design by Wiggins and McTighe.

UDL Guidelines: <https://udlguidelines.cast.org/>

TILT Framework: <https://www.tilthighered.com/>

Alternative Grading

Grading for Equity by Feldman: <https://crescendoedgroup.org/grading-for-equity/>

Grading for Growth by Clark and Talbert: <https://gradingforgrowth.com/>

Ungrading by Blum: <https://wvupressonline.com/ungrading>

JEDI Course Materials

<https://causeweb.org/jedi/>